



INTERNAL

Engineering Principles & Standards

Quenyx vOPS HUB — v1.0.0 RC1 · Document v2.0

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Version	Date	Notes
1.0	2026	Initial v1 pack (through Sprint 19).
2.0	2026-06-29	Aligned to v1.0.0 RC1 (Documentation Pack v2.0).

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Audience: The engineering org. **Purpose:** The non-negotiables that define how Quenyx is built. These are observed in the current codebase and are the bar for all future work.

1. Platform principles

- **One source of truth.** Operational + compliance data live in the platform; modules share identity, entitlements, audit, and the AI layer.
- **Thin edges, rich services.** Controllers validate and delegate; logic lives in `app/Services/**`.
- **Composable modules.** New modules plug in without re-plumbing identity/billing/AI.

2. No fake data

No `lorem`, `sample/demo/fake` controls, or placeholder business data in production paths. The corpus validator **rejects** fake-data markers. Executive metrics are **real counts**, never fabricated.

3. Source-of-truth database

The database is authoritative for business data. The AI core does **not** read business data directly — it consumes deterministic service outputs. Compliance content is imported from **official sources only**.

4. UUID-first

Domain entities (corpus, AI, RAG) use **UUIDs**. Public payloads expose UUIDs/codes — **never** raw auto-increment IDs.

5. Provenance-first

Every compliance element traces to a **source document** and an **import run**. Revisions are immutable snapshots. Nothing enters the corpus without provenance.

6. Fail-closed

Missing auth, missing entitlement, invalid input, or an uncitable AI answer → **reject**, don't degrade silently. Security and correctness defaults are restrictive.

7. Deterministic before probabilistic

A deterministic engine decides **what** is true and **what** to answer **before** any model is involved. Same inputs ⇒ same outputs. This is the core governance guarantee.

8. AI as renderer, platform as expert

The model **phrases**; it does not **decide**. Reasoning, gaps, recommendations, and citations come from deterministic engines. AI is off by default and provider-agnostic.

9. Module adapter pattern

Modules consume AI through `AiModuleAdapterInterface` (`buildContext` → `buildReasoning` → `buildPrompt`). Adapters **reuse** module services; they **never duplicate or move** business logic into the AI core.

10. Service boundaries

- Provider access only through `AiProviderRegistry` / provider classes. **No hardcoded models.**
- No direct OpenAI/HTTP calls outside provider classes (verified by static scan).
- DB access stays in the services/persistence layer, not the AI orchestration core.

11. Testing standards

- New services get deterministic unit tests; new routes get authorization/entitlement tests.
- Run `php artisan test (+ --filter=Compliance, --filter=AI)` before merge.
- Prefer deterministic fixtures over randomized data.

12. Documentation standards

- Docs live in `docs/quenyx-v1/` ; update them in the **same PR** as the code.
- Use the status legend (● / ● / ● / ●). **No fabrication.**
- Re-derive mechanical refs (Doc 08 routes, Doc 09 migrations) when they change.

13. Release standards

- New work is **feature-flagged off by default** when it touches AI or external calls.
- **Never edit applied migrations** — fix forward.
- Run the **Track-B audit** at phase boundaries; close high-risk findings before advancing.
- Keep `route:list` collision-free; keep `APP_DEBUG=false` in production.

14. Security-by-design

- Sanctum auth + project policy + module/QynShield entitlement on every tenant route.
- Audit sensitive actions.
- Prompt logging, conversation persistence, RAG, and tenant-evidence embeddings **off by default.**

- Secrets in env/secrets manager; `.env` never committed.
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The standard, in one line: *Deterministic, provenance-backed, fail-closed, UUID-first, AI-as-renderer — with no fake data and no fabricated claims, ever.*